

Education

Year	Degree/Certificate	Institute	CGPA
2026 - 2028	MSc in Human Computer Interaction	University of Siegen, Germany	-
2021 - 2025	BS in Artificial Intelligence	GIK Institute, Pakistan	3.47/4

Work Experience

Senior Technical Support Engineer

Motive Technologies, Inc.

Dec'25 - July'26
(US-based, remote)

- Resolved an average of 15 cases daily, diagnosing root cause within 1 to 2 hours and following up by phone and email, across client segments from SMB accounts to Strategic accounts with thousands of ELDs and vehicles.
- Maintained a 97% customer satisfaction rating in my final quarter and an average resolution time under 2 days, while keeping daily backlog under 15 open cases, and sustained these results even when case volume rose to 30 or more per day during a period when the team was understaffed.
- Led technical validation for AI-powered dashcams, troubleshooting root causes of AI-detected safety events and escalating verified issues to engineering, which directly informed product reliability and alert accuracy.
- Trained and ramped new hires to full case ownership in under 3 weeks, supporting team capacity during periods of high case volume.

Mitacs Globalink Research Intern – HCI and Generative AI

University of Regina, Prof Vanessa Figueiredo

Jun'25 - Sep'25
Regina, Saskatchewan, Canada

- Independently designed and built an intelligent tutoring agent that personalizes HCI academic content for undergraduate students based on individual cognitive ability and learning preference, reporting directly to the supervising professor.
- Built an automated RAG ingestion pipeline (n8n, Docker, Supabase, PostgreSQL) that pulls new files from Google Drive, extracts text across PDF, HTML, and OCR required formats, and chunks the content into Supabase, populated with a 44 chapter excerpt from The Encyclopedia of Human-Computer Interaction, 2nd Ed. as the core knowledge base.
- Architected a self-correcting, multi-agent answer pipeline: a generator agent drafted responses, an evaluator agent scored them against the source prompt, and a tester agent validated quality, automatically routing failed answers back to the generator until they passed.
- Designed a six-category test suite to evaluate RAG output quality, then benchmarked prompt variants against it: explicit structural rules produced more consistent formatting than few-shot examples, but a hallucination pattern, answering instead of declining when the material didn't cover the question, persisted regardless of prompt design.

Generative AI Research Internship

University of Europe for Applied Sciences, Prof Raja Hashim Ali

Jun'24 - Sep'24
Potsdam, Germany (Remote)

- Built a RAG based legal assistance system combining a quantized Llama 2 7B model with ChromaDB, evaluated across three German legal corpora, the Basic Law for the Federal Republic of Germany, Real Property Law, and the German Criminal Code, using LangChain for document loading, chunking, and retrieval.
- Benchmarked model outputs against human legal expert answers using BLEU and ROUGE metrics, achieving high scores on both, with an average response time of 72.36 seconds per query.
- Tested the system's ability to handle sensitive legal topics, including racism and terrorism, and to infer accurate answers to questions not explicitly stated in the source text, finding it performed reliably on both.
- Co-authored a manuscript based on this work as first and corresponding author, targeting Natural Language Processing (Cambridge University Press).

Infrastructure Operations/ Information Technology Internship

PTCL & Ufone4G

Jul'24 - Aug'24
Islamabad, Pakistan

- Independently deployed Red Hat OpenStack on both a laptop and a server as a proof of concept, evaluating deployment complexity as part of a feasibility study on whether the company should adopt it more broadly.
- Worked directly with Red Hat OpenStack, Red Hat Linux, Docker, and Kubernetes to assess containerization and orchestration performance across both environments.

Research Publications

The YOLOv8 Edge: Harnessing Custom Datasets for Superior Real-Time Detection (ICET'23)

doi.org/10.1109/ICET59753.2023.10374650

2023
IEEE Xplore

Conducted research on real time object detection using a custom trained YOLOv8 model for high accuracy and speed in video streams. Achieved an mAP50 of 0.864, demonstrating the effectiveness of domain specific datasets for robotics, autonomous vehicles, and video monitoring applications.

Influencing Factors in Facial Recognition and Verification: A Siamese Neural Network Hyperparameter Study (ICECT'24), First Author

doi.org/10.1109/ICECT61618.2024.10581383

2024
IEEE Xplore

- Performed a detailed study on optimizing Siamese neural networks for facial recognition, exploring hyperparameters such as optimizers, loss functions, and dataset diversity to enhance accuracy and performance in facial verification systems.

VGG Models in Image Captioning: Which Architecture Delivers Better Descriptions? (ICOSST'24), First Author 2025
doi.org/10.1109/ICOSST64562.2024.10871137 *IEEE Xplore*

- Conducted a comparative study of convolutional and LSTM architectures for image captioning, evaluating VGG16 and VGG19 models on the Flickr8k dataset using a custom Encoder Decoder architecture. Analyzed the effect of training duration on captioning quality and showed that VGG16 and VGG19 achieved comparable performance, offering insights into model convergence and evaluation benchmarks.

Academic Projects

Siamese Neural Network for Robust Facial Verification: Implementation & Hyperparameter Optimization Study

- Developed a Siamese Neural Network in TensorFlow for one-shot face verification, training on anchor-positive-negative triplets and evaluating performance using precision/recall metrics.
- Published research (ICECT'24) on optimizing hyperparameters (optimizers, loss functions, datasets), improving accuracy by 12% in low-data scenarios for facial recognition.
- Enhanced model robustness against lighting/pose variations using cross-dataset validation (LFW/AT&T), achieving near-supervised performance with minimal labeled data.

S.A.R.A - (Surgical Assistant Robotic Arm) - Bachelor Final Year Project

- Trained a YOLOv8 model to detect and classify surgical tools, using color coded sharpeners as a low cost proxy for real surgical instruments given hardware access constraints.
- Collected and annotated a training set of approximately 5,000 images across varied capture setups, then used annotation guided augmentation to expand the dataset and reduce overfitting.
- Contributed to the voice command pipeline (Whisper for speech to text, FLAN-T5 for intent parsing) that correctly resolved negated or multi part commands rather than relying on keyword matching alone.

E-Hospital: Multi-Role Healthcare Coordination App (Freelance Client Project)

- Designed and built the complete UI/UX for a healthcare coordination app as a solo freelance project, spanning 50+ screens across three roles, Doctor, Nurse and Patient, each with its own sign in flow and permission scoped interface.
- Modeled doctor and nurse workflows around a shared patient queue, viewing today's patients and each patient's details (name, age, presenting issue), with prescription writing scoped to doctors only.
- Built the patient facing booking flow, letting users schedule an appointment for themselves or someone else with a specific doctor, and view prescriptions from past visits.

Courses and Certifications

- **Courses:** - Machine Learning | Deep Learning | Natural Language Processing | Computer Vision | Data Engineering | Data Science | Cyber Security | Algorithms
- **Certifications:** - Finetuning Large Language Models (Lamini) | Foundational Generative AI (iNeuron.ai) | LLMOps (DeepLearningAI) | GenAI in Networking Bootcamp (Aviz Networks)

Awards and Achievements

- Awarded the **MITACS GRI 2025 Fellowship** for a fully funded research internship at the University of Regina, Canada
- Selected among 32 candidates from over 15,000 applicants for the **SummerSpark Internship Program** at PTCL & Ufone 4G
- Received the Dean's Honor Roll Award **5 times** during bachelor's studies
- Authored **3 peer-reviewed research publications** in reputable conferences
- **Winner, MLSA Hackathon, GIKI:** Built an online food ordering and delivery app in 3 days, replacing the university's existing over the counter ordering system
- **Microsoft Imagine Cup, Regional Finalist**, competing among top student innovators

Technical Expertise

- **Languages** - C | C++ | Python | Dart | SQL
- **Software and Libraries** - Claude Code | Jupyter Notebooks | NumPy | Pandas | OpenCV | Scikit-learn | SciPy | Pillow | Matplotlib | PyTorch | TensorFlow | Keras | Hugging Face Transformers | LangChain | Ollama | n8n | Docker | OpenStack | CTransformers | Whisper
- **Generative AI and LLMs** - LLMs | ChatGPT API | RAG (Retrieval-Augmented Generation) | Prompt Engineering | LLM Fine-tuning (LoRA, PEFT) | Local AI | Vector Databases (Supabase, Qdrant, ChromaDB) | Generative AI
- **Web and App Development** - Flutter | Firebase | React.js | REST APIs
- **Other Technical Skills** - CI/CD | Linux

Extra Curricular Activities

- Active member of **IEEE GIK Student Branch** for 4 years, progressing from volunteer to **General Secretary**, responsible for organizing technical events, workshops, and coordinating branch activities
- **Huawei Cloud Campus Coordinator for AI**, promoting Huawei Cloud at my university as the sole AI representative and arranging seminars and learning opportunities
- Core team member of the **Google Developers Student Club (GDSC)**, **GIKI**, dedicated to promoting Google technologies and hosting nationwide seminars and workshops